

Fatih Durmaz

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Education

MSc in IoT Security, Cyber Security (<i>Grande Distinction</i>) University of Southern Brittany & Université Libre de Bruxelles Dissertation: System Security Through Static Analysis by Detecting Dormant Vulnerabilities Advisors: Georgios Portokalidis, Jan Tobias Muehlberg	<i>Oct 2022 – Oct 2024</i> France & Belgium
BSc in Computer Science and Engineering (Gpa: 3.85/4.00) Sabancı University Dissertation: TimeInspector: A Static Analysis Approach for Detecting Timing Attacks Advisors: ErKay Savas, Cemal Yilmaz	<i>Oct 2017 – Jun 2022</i> Istanbul, Turkey

Research Experience

Research Intern , IMDEA Software Institute - Developed a methodology using static analysis to detect dormant vulnerabilities (<i>CVEs</i>) in systems under investigation, with a focus on Debian-based containers. - Designed and implemented a tool that integrates with containers to assess and notify whether a detected vulnerability poses a critical threat.	<i>Apr 2024 – present</i> Madrid, Spain
Software Development Intern , Nokia Bell Labs - Developed a state-of-the-art proof of concept (<i>PoC</i>) for ongoing research on multi-tenant client isolation in cutting-edge cloud FPGA environments. - Set up cloud deployment infrastructure using k3s and Docker. Documented <i>PoC</i> for the team. Tools: Golang, Kubernetes (K3s), Docker, Embedded Linux, Petalinux, Yocto	<i>Apr 2023 – Jul 2023</i> Lannion, France
Bachelor's Thesis , Sabancı University - Developed a static analysis method to detect binaries capable of executing timing attacks by analyzing control flow graphs and identifying suspicious execution paths. - Achieved high accuracy in detecting malicious binaries during experimental evaluations, with strong results across a range of benign and malicious samples.	<i>Oct 2022 – Jul 2023</i> Istanbul, Turkey
Undergraduate Researcher , Sabancı University - PURE 🔗 Solved matching problems using computational methods, focusing on bipartite graphs and perturbations for stable matching.	<i>Spring 2020</i> Istanbul, Turkey

Publications

TimeInspector: A Static Analysis Approach for Detecting Timing Attacks Fatih Durmaz, Musa Unal, Melih Taha Oz, Nureddin Kamadan, ErKay Savas, Cemal Yilmaz 10.1109/EuroSPW59978.2023.00037 🔗	<i>June 2023</i>
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Industry Experience

Software Developer Intern , Mainflux Upgraded InfluxDB to version 2 and migrated all queries from InfluxQL to FluxQL. Resolved issues with query comparators and added new comparison operators. Tools: Golang, Docker, nGinx, Go-Kit, InfluxDB, Microservices Architecture	<i>Feb 2022 – Oct 2022</i> Remote
Software Developer Intern , Tubitak Bilgem YTE (National Research Institute of Turkey) Built the frontend and backend for an event management and marketing platform. Tools: Spring Boot, React.js, PostgreSQL, Spring Security	<i>Aug 2021 – Oct 2021</i> Istanbul, Turkey

Data Science Intern, Faraday Networks

- Developed a tool for anomaly detection in time-series data of customer Wi-Fi sessions, using machine learning models to identify abnormalities.
Tools: PostgreSQL, Python, Facebook Prophet

Jun 2020 – Aug 2020
Istanbul, Turkey

Teaching Experience

Student Assistant, Sabanci University - Advanced Programming (CS 204)

- Created lecture slides on Smart Pointers in C++ and prepared weekly quizzes to reinforce key concepts.

Fall 2019
Lannion, France

Selected Projects

Risk Analysis

- Analyzed and presented NIST SP 800-39 Standard on 'Managing Information Security Risk.'
- Led a team of 10 and employed peer review processes to ensure high-quality deliverables.

Nov 2022
Lorient, France

CSPN Evaluation of Signal App

- Conducted a CSPN Target Evaluation of the Signal app as part of a course on secure advanced programming. Assessed security protocols and data protection measures.

Oct 2022 – Dec 2022
Lorient, France

Lightweight Cryptography Analysis

- Analyzed the Simon and Speck whitepaper for lightweight cryptography, implemented the algorithms, and demonstrated performance differences in software.

Oct 2022 – Nov 2022
Lorient, France

Skills

Languages: English (IELTS 7.5/9), Turkish (Native), French (A2 ULB)

Programming Languages: Python***, C++***, Golang**, C**, Java*, Rust*

Honors and Distinctions

2024 Master's in IoT Security (UBS & ULB)

Honors (*UBS*) and Grande Distinction (*ULB*)

2023 Erasmus+ European Commission

Full Erasmus Mundus Scholarship For Masters Program

2022 Sabanci University

Ranked top 4% in Natural Sciences & Engineering Faculty

2022 Sabanci University

Dean's Honor List

2017–2022 Sabanci University

Dean's High Honor List (x5)

2017 National University Admission Exam

Ranked 1481st (top 0.07%)